CTS — Aptitude Test 4

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Question 76. Without stoppages a train travels a certain distance with an average speed of 80 km/h and with stoppages with an average speed of 60 km/h.

The time is increased from 10:00 to 10:30.
That means the increased time 30 minutes is half of the actual time taken to travel.
Actual time taken to travel = 30 x 2 = 60 minutes.
Raghav starts at home everyday at 9:00am.
The distance from his home to office = 1 x 15 = 15 km.
Time taken to reach the office at 10:15 = 1.25 hours
Speed required to cover 15km in 1.25 hours = 15/1.25 = 12 kmph

How many minutes per hour does the train stops?

Solution:
Distance travelled by the train without stoppages in one hour is 80km.
Even when the train is travelling with stoppages, the speed of the train is 80kmph.
Time taken for the train to travel 60km at the speed of 80kmph = 60/80 = 0.75 hour = 45minutes.
The train has travelled for 45 minutes.
That means the stoppages are for 60 – 45 = 15 minutes.

Question 77. A man covers a distance of 1200 km in 70 days resting 9 hours a day. If he rests 10 hours a day and walks with speed 1½ times of the previous in how many days will he cover 750 km?

Solution:
To cover 80km in 8 hours, the average speed required is 10kmph.
In the question, the two speeds given are 6kmph and 8kmph.

For these two values an average speed of 10kmph can never be achieved.
This is one of the dummy question, which must be avoided in the examination.

Solution:
Total time taken to cross 1200 km in the first case = 70 x (24 – 9) = 1050 hours
Speed of the man = 1200/1050
= 8/7 kmph
1 ½ times the original speed = 1.5 x 8/7
= 12/7 kmph.
Time taken to cross 750 km at a speed of 12/7kmph = 750/(12/7)
= 437.5 hours.
If the man is taking rest for 10 hours a day in the second case, he will be walking (24 – 10) = 14 hours a day.
Number of days taken to cover the 750km distance = 437.5/14
= 31.25 days.

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Question 78. If Raghav travels at a speed of 15kmph he reaches office at 10am. If he travels at 10kmph he reaches office at 10:30 am. At what speed should he travel so that he reaches office at 10:15am? (assume that he leave home at same time and takes the same route).

Solution:
The distance in each case is same and there are four cases.
Assume that the speed in each case is
S1 = 10kmph; S2 = 20kmph
S3 = 30kmph; S4 = 40kmph

4

Solution:
When the speed is decreased by 1/3rd, the time will increase by 1/2.

As

By solving, Average speed As=19.2kmph

S2
+
+
1

+
=
1

1

1

S4

S3

S1

Question 79. A man decided to cover a distance of 80km in 8 hrs. He can walk at a speed of 6 kmph and cycle at 8 kmph successively. How long will he walk and how long will he cycle?

Question 80. Megha drives along the perimeter of square field of side 10 km. She drives along the first side at 10 kmph, along the second side at 20 kmph, along the third side at 30 kmph and along the fourth side at 40 kmph. Her average speed is?

Question 81. A goods train leaves a station at a certain time and at fixed speed. After 6hrs another train leaves the same station and move in same direction at an uniform speed of 90kmph. The train catches up the goods train in 4hrs. Find the speed of the goods train.

Solution:
The distance traveled by the second train to catch the first train in 4 hours = 4 x 90 = 360 km.
The distance travelled by the goods train = 360 km.
Time taken by goods train to travel 360km = 6 + 4 = 10 hours.
Speed of goods train = 360/10

= 36 kmph.

Question 82. Five different roads join a village to the nearby city. The number of different ways in which a person can go to the town and come back is?

Solution:
Number of ways to go from village to city = 5
Number of ways to return to the village = 5
The two events are dependent with each other.
For dependent events, we have to multiply.
Number of ways = 5 x 5 = 25.

Question 83. Using the digits 1,2,3,4,5,6 and 7 how many 4 digit number can be formed without repetition?

Solution:
There are 7 elements out of which 4 elements should be taken and arranged.
Number of ways nPr = 7P3
= 7 x 6 x 5 = 210

Question 84. What is the sum of all four digit numbers that can be formed using the digits 1, 2, 5, 8 without repetition?

Solution:

The number of four digit numbers that can be formed using 1, 2, 5 and 8
= 4! = 24.
Among the 24 numbers, in the units digit, each of the four numbers will occur for 24/4 = 6 times.
Sum of all the unit digits
= 24(1+2+5+8) = 384
Sum of all tens place numbers
= 3840
Sum of all 100's place numbers
= 38400
Sum of all thousands place numbers
= 384000
Sum of all 24 different numbers
= 384 + 3840 + 38400 + 384000
= 426624

Question 85. Using the digits 2, 3, 6 and 7 how many four digit numbers can be formed without repetition, that are divisible by 4?

Solution:
Condition for divisibility by 4 is, the last two digits of the number must be divisible by 4.
The different pairs that can be formed using the given numbers which are divisible by 4 are:
32, 36, 72, 76.
There are four cases and in each case there will be 2 more numbers left to be arranged. They can be arranged in 2! Ways.
Total possible ways = 4 x 2! = 8

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Question 86. How may 5 digit odd numbers can be formed from 1 2 3 4 5 without repeating any numbers?

Solution:
To be an odd number, the last digit must be an odd number.
The last digit can be either 1 or 3 or 5.
In each case the other four places can be arranged in
4! Ways.
Total possible ways
= 3 x 4! = 72

Question 87. How many number of three digit numbers can be formed using the numbers 2,3,4,5 with no repetition?

Solution:
4P3 = 4 x 3 x 2 = 24 number of four digit numbers can be formed.

Solution:
Probability
of A and B completing the
test on time and C not completing on
time
= (1/2)(2/3)(2/5) = 4/30
Probability
of B and C completing the

test on time and A not completing on
time
= (2/3)(3/5)(1/2) = 1/5
Probability
of A and C completing the
test on time and B not completing on
time
= (1/2)(3/5)(1/3) = 1/10
Probability
of
only
two
of
them
completing the test on time
= (4/30) + (1/5) + (1/10)
= 13/30

Question 88. From the cards Jack, Queen, King and Ace are removed. The algebraic sum of the rest of cards will be?

Solution:
After
removing
the
face
cards,
the
remaining cards will be numbered from
2 to 10.
Sum of the numbers from 2 to 10
= 54.
Sum of all the number cards
= 4 x 54 = 216

Question 89. What is value of 15C13 ?

Solution:
nCr = nC(n-r)
So, 15C13 = 15C15-13
= 15C2
= (15 x 14)/(2 x 1) = 105

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Question 90. If nC5=nC6, what is the value of n?

Solution:
Probability of guessing correctly for one
time = ½.
Probability of guessing correctly for
three times in a row
= ½ x ½ x ½ = 1/8

Solution:
nCr = nC(n-r)
nC5 = nC6

r = 5 and n –r = 6
n – 5 = 6
n = 11

Question 91. In how many ways can the captain of a cricket team select 11 players from a squad of 14 players?

Solution:
Since the person selecting is the captain,
he will definitely get selected.
So
the
captain
has
to
select
the
remaining
10
players
from
the
remaining 13 players.
Ways of selecting = 13C10
= (13 x 12 x 11)/(3 x 2 x 1) = 286 ways.

Solution:
Probability of getting at least one Nestle
= 1 – Probability of getting no Nestle
Probability of getting no Nestle
= 5C2/15C2 = 10/105
Probability of getting at least one Nestle
= 1 – (10/105) = 95/105 = 19/21

Question 92. The Probability of finishing a test on time by A is 1/2, B is 2/3 and by C is 3/5. If all of them write the test independently, then what is the probability that just two of them are able to write the test on time?

Question 93. Varun is guessing which of the 2 hands holds a coin. What is the probability that Varun guesses correctly three times in a row?

Question 94. Ravi has a bag of 10 Nestle and 5 Cadbury chocolates. Out of these, he draws two chocolates. What is the probability that he would get at least one Nestle Chocolate?

Question 95. Two distinct no's are taken from 1,2,3,4.....28. Find the probability that their sum is less than 13.

Solution:
Let us assume that the two numbers to be chosen are x and y.
Given, x + y < 13
The maximum possible value for x + y = 12
If x = 1, then y can be any value from 2 to 11. There are 10 pairs if x = 1
If x = 2, then y can be any value from 1 to 10 except 2. There are 9 pairs if x = 2
If x = 3, then y can be any value from 1 to 9 except 3. There are 8 pairs if x = 3
Similarly when x = 4 there are 7 pairs
When x = 5 there are 6 pairs
When x = 6 there are 5 pairs
When x = 7 there are 5 pairs
When x = 8 there are 4 pairs
When x = 9 there are 3 pairs
When x = 10 there are 2 pairs and
When x = 11 there is 1 pair.
Total number of ways in which there will be a sum less than 13 is 60.
Total number of ways in which two numbers can be selected from1 to 28 is
= 28C2 = (28 x 27)/(2 x 1) = 378.
Probability = 60/378

Question 96. There are 5 dogs and three cats. What is the probability that there is one cat at both the ends when arranged?

Solution:
Number of ways of selecting two cats to sit at the end = 3C2 = 3 ways.
Out of the three different cases in all cases the remaining 5 dogs and 1 cat can be arranged in 6! ways.
The two cats which are at the end can also be rearranged in 2! ways.
Total number of arrangements in which the two cat sits at the each end
= 2! x 3 x 6!
= 4320 ways.
Total number of ways arranging 5 dogs and three cats = 8! = 40320
Probability = 4320/40320

Question 97. Two cards are drawn at random from a pack of 52 cards. What is the probability that either both are black or both are queen?

Solution:
Number of ways of selecting two black cards = 26C2 = 325 ways.
Number of ways of selecting two queen cards = 4C2 = 6 ways.
Total ways of selecting two black cards or two queens = 325 + 6 – 1 = 330.
We are subtracting 1 because out of the six ways of selecting 2 queens in one way, both the queen cards will be black.
Total ways of selecting 2 cards out of 52 cards = 52C2 = 1326
Probability = 330/1326

Question 98. Dividend of Rs. 504 lakhs for shares was announced by a company. 100 employee cum share holders get Rs. 3.60 lakh each & the share holder who is not the employee gets Rs. 2.40 lakh each. How many share holders are there who are not employee?

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Solution:
Number of share holders who are employees = 100.
Assume that the

number
of
share
holders who are not employees = N.
100 x 3.6 + N x 2.4 = 504
2.4N = 504 – 360
2.4N = 144
N = 60
There are 60 shareholders who are not
employee of the company.

Question 99. If f(X)= 2X-1 + f(X-1), if X is not equal to zero and if f(X=0)=0, find the value of f(5).

Solution:
f(5) = 2(5) – 1 + f(4)
f(4) = 2(4) – 1 + f(3)
f(3) = 2(3) – 1 + f(2)
f(2) = 2(2) – 1 + f(1)
f(1) = 2(1) – 1 + f(0)
f(1) = 1

f(2) = 2(2) – 1 + 1 = 4
f(3) = 2(3) – 1 + 4 = 9
f(4) = 2(4) – 1 + 9 = 16
f(5) = 2(5) – 1 + 16 = 25

Question 100. APPLE + PEAR = GRAPE. G+R+A+P+E = ?

Solution:
The problem is based on
Crypt arithmetic.
Each letter is assigned with a distinct
one digit number.
The different letter in the question are
A, P, L, E, R and G

A P P L E

+ P E A R

G R A P E

A + 1 = G
E + R = E, So R = 0
P + P = R ■P + P = 0, So P = 5

C T S

P R A G L
E

5 0 1 2 4
6
X

5 0 2 3 3
X

5 0 3 4 2
8
X

5 0 4 5
X

5 0 5
X

5 0 6 7 9 10 X

5 0 7 8 8
X

5 0 8 9 7
2
√

P = 5; R = 0; A = 8; G = 9; L = 7 and
E = 2
G+R+A+P+E = 9+0+8+5+2 = 24

There are two C’s in
the company’s logo.

Blue 'C' represents
the 'Customer' and
Green 'C' represents

the 'Company' .

It symbolizes that

the company is
'Flexible' for the
customer. Both 'C's

are over-laced
symbolizing the
'bond' and close

interaction.